

Problem 10.34

You are given that

$$\delta_t = \frac{0.2t}{1 + 0.1t^2}.$$

Calculate i_2 .

Solution

The accumulation function is related to the force of interest by

$$a(t) = \exp \left(\int_0^t \delta_x dx \right).$$

Substituting the given force of interest,

$$a(t) = \exp \left(\int_0^t \frac{0.2x}{1 + 0.1x^2} dx \right).$$

Since

$$\frac{d}{dx} \ln(1 + 0.1x^2) = \frac{0.2x}{1 + 0.1x^2},$$

we have

$$\begin{aligned} a(t) &= \exp \left(\ln(1 + 0.1x^2) \Big|_0^t \right) \\ &= \exp(\ln(1 + 0.1t^2)) \\ &= 1 + 0.1t^2. \end{aligned}$$

Thus,

$$a(1) = 1.1, \quad a(2) = 1.4.$$

The effective interest rate during the second period is

$$\begin{aligned} i_2 &= \frac{a(2)}{a(1)} - 1 \\ &= \frac{1.4}{1.1} - 1 \\ &= \frac{1.4 - 1.1}{1.1} \\ &= \frac{3}{11}. \end{aligned}$$

Therefore,

$i_2 = \frac{3}{11} \approx 0.2727 = 27.27\%.$
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